Week 13: Decisions, Games, and Uncertainty

Pareto Optimization, Prisoner's Dilemma, Probability, and Statistical Inference

Instructor / Mentor

Kiki Research Training Program

Week 13

Contents

Class Mission

The Question That Connects Today

Optimization works when outcomes are known

Convert the important factors into a scalar score, compare the scores, and choose the smallest or largest one.

But what if the outcome is random?

If X and Y are random variables, what does it mean to say that one is "greater" than the other?

many objectives → one scalar score → many players → random outcomes

Week 13 Mission

1. Finish multi-objective optimization using dominance and Pareto frontiers.
2. Finish game theory using best responses and the prisoner's dilemma.
3. Compare uncertain choices using probability distributions, expectation, and risk.
4. Use data to test a claim with a null hypothesis, p-value, and significance level.
5. Connect statistical inference to the design of scientific experiments.

Today's promise

By the end, students can explain why "best," "stable," and "supported by evidence" are three different ideas.

120-Minute Flow

1. **0–25 min:** multi-objective and Pareto optimization.
2. **25–50 min:** prisoner's dilemma and Nash equilibrium.
3. **50–55 min:** break and transition question.
4. **55–80 min:** basic probability and random variables.
5. **80–105 min:** hypothesis tests, p-values, and significance.
6. **105–115 min:** Python simulation and scientific-experiment design.
7. **115–120 min:** exit ticket and homework briefing.

Finish Multi-Objective Optimization

Recall: One Objective Gives One Ranking

Suppose a grid operator chooses a plan x and minimizes cost:

$$\min_x f(x) = \text{cost}(x).$$

- Every feasible plan receives one number.
- Smaller numbers rank ahead of larger numbers.
- The objective converts a complicated decision into a comparison.

Example

If plans A, B, and C cost $\$35$, $\$50$, and $\$80$, cost alone selects A.

Two Objectives May Disagree

Lower cost and lower emissions are both better.

Plan	Cost	Emissions
A: fossil-heavy	35	95
B: balanced	50	60
C: renewable-heavy	80	25
D: inefficient backup	90	90

Think–pair–share

Which plan is best? What extra information do you need before answering?

Dominance: A Comparison That Needs No Weights

Plan u **dominates** plan v when:

1. u is no worse than v in every objective, and
2. u is strictly better in at least one objective.

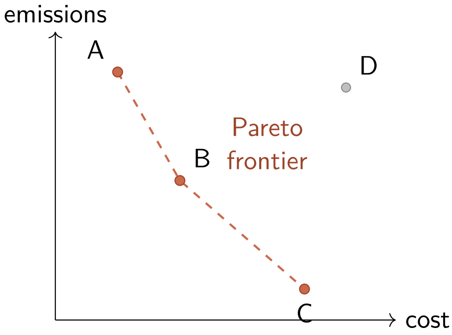
Apply the rule

B has cost $50 < 90$ and emissions $60 < 90$, so B dominates D. There is no reason to choose D from these two objectives alone.

A and C do not dominate each other

A is cheaper, but C has lower emissions. The objectives disagree.

Pareto Frontier: The Surviving Tradeoffs



Pareto-optimal

A plan is Pareto-optimal if no available plan dominates it.

- Frontier: A, B, C.
- Dominated: D.
- A frontier is a menu, not one winner.

Weighted Sum: Convert Two Goals to One Score

$$J = \lambda(\text{cost}) + (1-\lambda)(\text{emissions}), \qquad 0 \le \lambda \le 1.$$

λ	What receives more weight?	Selected plan
0.2	emissions	C: $0.2(80) + 0.8(25) = 36$
0.6	more weight on cost	B: $0.6(50) + 0.4(60) = 54$
0.8	cost	A: $0.8(35) + 0.2(95) = 47$

Interpretation

The weight does not discover our values. It **encodes** the values we choose.

Important Pitfall: Units Can Secretly Choose the Winner

What if cost is measured in dollars and emissions in kilograms?

$$J = 0.5(\$50{,}000) + 0.5(60 \text{ kg})$$

The cost number dominates simply because its numerical scale is larger.

Normalize before weighting

One simple transformation is

$$z = \frac{x - x_{\min}}{x_{\max} - x_{\min}},$$

which puts each objective on a comparable 0–1 scale.

Modeling lesson

A scalar objective is useful only when its units, scaling, and weights have a defensible meaning.

Pareto Decision Checklist

1. State every objective and whether to minimize or maximize it.
2. Remove infeasible plans.
3. Remove dominated plans.
4. Display the Pareto frontier.
5. Normalize objectives if their units or scales differ.
6. Choose a frontier point using values, constraints, or stakeholder priorities.
7. Test whether the choice changes when assumptions or weights change.

The frontier improves the conversation

Instead of arguing about one mysterious score, decision-makers can see exactly what must be sacrificed to improve another goal.

Finish Game Theory

From One Decision-Maker to Several

Optimization

One decision-maker chooses x to improve one objective:

$$\min_x f(x).$$

Game theory

Each player chooses a strategy, but each payoff depends on **everyone's** strategies:

$$u_i(s_i, s_{-i}).$$

New difficulty

My best action can change when your action changes.

Four Pieces of a Game

1. **Players:** who makes decisions?
2. **Strategies:** what can each player choose?
3. **Payoffs:** how much does each player value every outcome?
4. **Information:** what does each player know when choosing?

Two-household grid game

Players are households A and B; each can shift usage or use power at the peak; payoffs combine comfort, price, and rewards.

Prisoner's Dilemma: The Story

Two prisoners are questioned separately. Each chooses to **stay silent** or **confess**. Payoffs are negative years in prison, so larger is better.

	B stays silent	B confesses
A stays silent	$(-1, -1)$	$(-5, 0)$
A confesses	$(0, -5)$	$(-3, -3)$

Read one cell

If A stays silent and B confesses, A receives -5 and B receives 0.

Best Response: Hold the Other Choice Fixed

A's best responses:

- If B stays silent, A compares -1 with 0: **confess**.
- If B confesses, A compares -5 with -3: **confess**.

B's payoff matrix is symmetric, so B reaches the same conclusion.

Dominant strategy

A strategy is dominant if it is a best response to every strategy of the other player. Confess is dominant for both prisoners.

Nash Equilibrium: Two Best Responses Meet

A strategy pair (s_A^*, s_B^*) is a Nash equilibrium if neither player can improve by changing alone:

$$u_A(s_A^*, s_B^*) \geq u_A(s_A, s_B^*) \quad \text{and} \quad u_B(s_A^*, s_B^*) \geq u_B(s_A^*, s_B).$$

Prisoner's dilemma equilibrium

At (confess, confess), either player who switches alone changes from -3 to -5. Therefore no unilateral switch helps.

Nash does not mean morally good, fair, or globally best

It means stable against one player's unilateral deviation.

Nash Equilibrium vs. Pareto Improvement

Compare two outcomes:

Outcome	A payoff	B payoff
Both confess	-3	-3
Both stay silent	-1	-1

- Both staying silent makes both players better off.
- Thus (confess, confess) is Pareto dominated by (silent, silent).
- Yet (silent, silent) is unstable: either prisoner can gain by confessing alone.

Central lesson

Individually rational choices can create a collectively worse result.

Smart-Grid Prisoner's Dilemma

Two households decide whether to shift flexible demand away from the system peak.

	B shifts	B uses peak
A shifts	$(3, 3)$	$(1, 4)$
A uses peak	$(4, 1)$	$(2, 2)$

1. Find A's best response in each column.
2. Find B's best response in each row.
3. Locate the Nash equilibrium.
4. Identify the Pareto improvement.

Mechanism Design: Change the Payoffs

Suppose the operator adds a two-point reward for shifting.

	B shifts	B uses peak
A shifts	$(5, 5)$	$(3, 4)$
A uses peak	$(4, 3)$	$(2, 2)$

What changed?

Shifting is now each household's best response in both cases, so (shift, shift) becomes the Nash equilibrium.

Engineering viewpoint

Do not merely predict behavior. Design prices or rules so individually attractive behavior also helps the grid.

From Scores to Random Variables

Transition Question: Which Random Cost Is Smaller?

Plan X uses uncertain renewable power; plan Y buys a fixed contract.

Plan	Possible cost	Probability	Expected cost
X	20 or 80	0.60 or 0.40	$0.6(20) + 0.4(80) = 44$
Y	always 45	1.00	45

Can we simply say $X < Y$?

X has lower expected cost, but it costs more than Y in 40% of cases. The answer depends on what kind of comparison matters.

Probability Gives Several Valid Comparisons

For random variables X and Y, we might compare:

- **Expected value:** Is $E[X] < E[Y]$?
- **Chance of winning:** Is $P(X < Y) > 0.5$?
- **Risk:** Which has smaller variance or fewer extreme outcomes?
- **Threshold reliability:** Is $P(X \leq 50) \geq 0.95$?
- **Worst case:** What is the largest plausible loss?

No universal meaning of "greater"

A random variable is a distribution of possible values, not one known scalar. We must state the comparison rule.

Probability Foundations

Probability Starts with Outcomes and Events

Sample space Ω

The set of all possible outcomes. For two coin flips,

$$\Omega = \{HH, HT, TH, TT\}.$$

Event

A set of outcomes. “At least one head” is

$$A = \{HH, HT, TH\}.$$

If the four outcomes are equally likely, $P(A) = 3/4$.

Three Basic Probability Rules

1. Bounds: $0 \leq P(A) \leq 1$.
2. Certainty: $P(\Omega) = 1$.
3. Complement: $P(A^c) = 1 - P(A)$.

For any two events,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Grid example

If $P(\text{overload}) = 0.10$, then $P(\text{no overload}) = 0.90$.

Conditional Probability: Information Changes Chances

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B) > 0.$$

Hot-day overload example

Out of 100 days, 20 are hot and 8 are both hot and overloaded:

$$P(\text{overload} \mid \text{hot}) = \frac{8}{20} = 0.40.$$

If 10 of all 100 days overload, then $P(\text{overload}) = 0.10$.

Interpretation

Knowing that a day is hot changes the relevant comparison group from all days to hot days.

Independence Is a Strong Claim

Events A and B are independent when

$$P(A \cap B) = P(A)P(B),$$

or equivalently $P(A \mid B) = P(A)$ when probabilities are nonzero.

- Separate fair coin flips are commonly modeled as independent.
- Temperature and air-conditioning load are not independent.
- "Different variables" does not imply "independent variables."

Random Variable: Assign a Number to Each Outcome

A random variable is a function from outcomes to numbers:

$$X : \Omega \to \mathbb{R}.$$

Two coin flips

Let X be the number of heads.

Outcome	HH	HT, TH	TT
X	2	1	0
$P(X = x)$	$1/4$	$1/2$	$1/4$

The distribution tells us which values are possible and how likely they are.

Expectation and Variance Summarize a Distribution

For a discrete random variable,

$$E[X] = \sum_x x P(X = x), \qquad \mathrm{Var}(X) = \sum_x (x - E[X])^2 P(X = x).$$

- Expected value describes the long-run average or center.
- Variance describes squared spread around that center.
- Standard deviation $\sigma = \sqrt{\mathrm{Var}(X)}$ uses the original units.

Expectation is not a guaranteed outcome

The expected value of one fair die roll is 3.5, even though 3.5 cannot appear on the die.

Return to Plans X and Y

Plan X

$$E[X] = 44$$

$$\mathrm{Var}(X) = 0.6(20-44)^2 + 0.4(80-44)^2 = 864$$

Thus $\sigma_X \approx 29.4$.

Plan Y

$$E[Y] = 45, \qquad \mathrm{Var}(Y) = 0.$$

Also,

$$P(X < Y) = 0.60.$$

Decision

X wins on expected cost; Y wins on predictability. The preferred plan depends on risk tolerance and reliability requirements.

From Probability to Statistical Inference

Probability Goes Forward; Statistics Reasons Backward

Probability

Assume a model is known. Predict what data could occur.

model $\longrightarrow$ possible data

Statistical inference

Observe a sample. Evaluate claims about the unknown process.

observed data $\longrightarrow$ evidence about model

Research question

Is a new fault detector genuinely better, or could its observed improvement be random sampling variation?

Population, Sample, Parameter, Statistic

Population	Every case we want to understand.
Sample	The cases we actually observe.
Parameter	An unknown population quantity, such as true accuracy p.
Statistic	A number calculated from the sample, such as sample accuracy $\hat{p}$.

Sampling variation

Two random samples from the same population usually produce different statistics. Inference asks whether a difference is larger than ordinary random variation.

Start with Two Competing Hypotheses

Suppose we test whether a coin favors heads.

Null hypothesis

$H_0 : p = 0.5$. The coin is fair; observed imbalance is sampling variation.

Alternative hypothesis

$H_1 : p > 0.5$. The coin favors heads.

We flip the coin 10 times and observe 9 heads.

Why write hypotheses first?

The direction of the claim determines which outcomes count as "at least as extreme."

What a p-Value Actually Asks

Definition in words

Assuming H_0 is true, the p-value is the probability of data at least as extreme as what we observed, using the chosen test statistic.

For the one-sided test with 9 heads out of 10,

$$p\text{-value} = P(X \geq 9 \mid X \sim \text{Binomial}(10, 0.5))$$

$$= \frac{\binom{10}{9} + \binom{10}{10}}{2^{10}} = \frac{11}{1024} \approx 0.0107.$$

It is not $P(H_0 \text{ is true} \mid \text{data})$

A p-value treats the null hypothesis as an assumption; it does not report the probability that the hypothesis is true.

Significance Level: Make the Decision Rule in Advance

Choose α before examining the result; a common classroom value is 0.05.

Result	Decision
p-value $\leq \alpha$	Reject H_0
p-value $> \alpha$	Fail to reject H_0

Here $0.0107 < 0.05$, so we reject H_0 at the 5% level.

Precise language

"Fail to reject" does not mean "prove" or "accept" the null. The data may simply be insufficient.

One-Sided and Two-Sided Questions Differ

- One-sided: Does the coin favor heads? $H_1 : p > 0.5$.
- Two-sided: Is the coin unfair in either direction? $H_1 : p \neq 0.5$.

For 9 heads in 10 flips, the symmetric two-sided p-value is

$$P(X \geq 9) + P(X \leq 1) = \frac{22}{1024} \approx 0.0215.$$

Research integrity

Choose the hypothesis and analysis plan before seeing the data. Changing the question afterward can make evidence look stronger than it is.

Two Ways a Hypothesis Test Can Be Wrong

	H_0 true	H_0 false
Reject H_0	Type I error	Correct detection
Fail to reject H_0	Correct restraint	Type II error

- Type I error: false positive; its long-run rate is controlled by α under the test assumptions.
- Type II error: false negative; its probability is often written β.
- Power $= 1 - \beta$: chance to detect a real effect of a specified size.

Statistical Significance Is Not Practical Importance

- A tiny effect can have a small p-value with a huge sample.
- An important effect can miss significance with a small, noisy sample.
- A p-value does not measure effect size, engineering value, or data quality.

Report more than a p-value

Include the estimated effect, uncertainty such as a confidence interval, sample size, assumptions, and practical consequences.

Grid example

A controller may reduce peak load by a statistically detectable 0.1% but still not save enough money to justify deployment.

Experiments and Simulation

Statistics Is Built on Experimental Design

A strong experiment uses:

1. **Comparison:** treatment group versus a meaningful control.
2. **Random assignment:** reduces systematic group differences.
3. **Replication:** enough independent observations to estimate variation.
4. **Blinding when possible:** reduces measurement and expectation bias.
5. **Pre-specified outcomes:** prevents selecting only favorable results.

A small p-value cannot repair a biased experiment

Statistical formulas rely on how the data were produced. Bad sampling or confounding can invalidate the conclusion.

Example Experiment: Demand-Response Message

Question

Does a new text message reduce household peak demand?

1. Recruit eligible households and record a baseline period.
2. Randomly assign households to the new message or standard message.
3. Define the primary outcome: average kW during 5–8 p.m.
4. State H_0: no difference in mean peak demand.
5. Estimate the effect and its uncertainty; compute the planned test.
6. Check practical size, missing data, assumptions, and side effects.

Python Lab: Compare Random Costs

```python
import numpy as np

rng = np.random.default_rng(13)
n = 100_000

# Plan X costs 20 with probability 0.6, otherwise 80.
X = rng.choice([20.0, 80.0], size=n, p=[0.6, 0.4])
Y = np.full(n, 45.0)

print("mean costs:", X.mean(), Y.mean())
print("standard deviations:", X.std(), Y.std())
print("P(X < Y):", np.mean(X < Y))
print("P(X > 50):", np.mean(X > 50))
```

Simulation idea

Repeated random trials approximate probabilities through long-run relative frequencies.

Python Lab: Simulate a p-Value

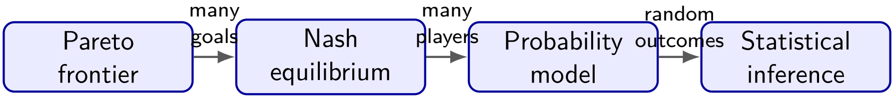
```python
rng = np.random.default_rng(13)
n_simulations = 100_000

# H0: a fair coin. Count heads in 10 flips.
null_counts = rng.binomial(n=10, p=0.5, size=n_simulations)
observed_heads = 9

# One-sided alternative: the coin favors heads.
p_value = np.mean(null_counts >= observed_heads)
print("simulated p-value:", p_value)
print("exact p-value:", 11 / 1024)
```

Monte Carlo uncertainty

The simulated answer changes slightly across runs. More simulations usually make it more stable.

Wrap-Up and Homework

The Complete Map



- Pareto analysis asks which tradeoffs are undominated.
- Nash analysis asks which strategic choices are stable.
- Probability describes uncertain outcomes under a model.
- Inference asks what observed data say about an unknown process.

Exit Ticket

Answer in one sentence or one calculation each:

1. Why is a Pareto frontier usually a set rather than one answer?
2. Why is (confess, confess) stable but collectively worse?
3. Give two different ways to compare random variables X and Y.
4. What does a p-value assume is true?
5. Why do we say "fail to reject" instead of "accept" H_0?

Homework for This Week

Manual programming task

Complete `week13_probability_games.py`: Pareto filtering, pure Nash equilibrium detection, random-cost comparison, and a simulated one-sided p-value.

Reasoning task

Analyze a demand-response game and design a controlled experiment for testing a new grid intervention.

AI policy

AI may explain concepts or help interpret an error message, but students must write the required TODO logic and evidence-based conclusion themselves.

Key Takeaways

1. Scalar objectives rank choices; Pareto frontiers preserve meaningful tradeoffs.
2. Nash equilibrium means no player benefits by switching alone, not that the outcome is socially best.
3. Random variables require distributional comparisons such as expectation, variance, and probability of exceeding a threshold.
4. A p-value measures how surprising the data are under a stated null model.
5. Scientific evidence depends on experimental design, effect size, uncertainty, and assumptions—not a p-value alone.